Magnitude and Flux Density Variations of RU Camelopardalis

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Abstract

RU Camelopardalis, also known as RU Cam, is a Type II Cepheid variable star. We determine that the magnitude and flux density of RU Cam in the V-band fluctuate over a 21 day period of observation. RU CAM varied between magnitudes of 7.7 and 9.9. RU CAM's flux density varied between 4.09×10^{-28} W m^{-2} Hz^{-1} and 2.97×10^{-27} W m^{-2} Hz^{-1}. Along with the plotted variability over time, the uncertainty due to photon and readout noise were also visualized. Using Python, SAO DS9, and astrophotometry, I analyzed the images and determined the aforementioned values.

1 Introduction

RU Camelopardalis, otherwise known as RU Cam, is a Type II Cepheid variable star of the W Virginis variable subgroup in the constellation Camelopardalis.[Kollath 1993] It has an average period of 22 days, Kollath states. According to the SIMBAD astronomical database, RU Cam has a right ascension of $07^h21^m44^s$ and a declination of $+69^o$ 40' 15". [Wenger 2000] Kipper [2007] states RU Cam is a carbon star. Its atmosphere contains more carbon than it does oxygen. Kipper [2007] notes that there are not many Type II Cepheid stars that are also carbon stars, making RU Cam an unusual case. RU Cam was discovered in 1907 by Ceraski and was subsequently reported to be a new variable star with its distinction as a Type II Cepheid following soon after. [Ceraski 1907]

Type II Cepheid variables are stars with varying periods between one and 50 days. Variable stars are a type of star whose apparent magnitude – brightness seen from Earth – changes over a given period of time. [Wallerstein 2002] Type II Cepheids are typically found in older stellar populations with a low metallicity, Das [2026]states. RU Cam is a W Virginis variable. [Kollath 1993] Type II Cepheids are commonly divided into three subclasses depending on their periods: BL Herculis variables (1-5 days), W Virginis variables (10-20 days), and RV Tauri variables (periods greater than 20 days). [Wallerstein 2002] While RU Cam exceeds the typical period constraints of a W Virginis variable, it is commonly referred to as such in publications.

Cepheid variables are useful as distance indicators, or standard candles, because of their period-luminosity relationship. [Das 2026] The period-relationship describes the directly proportional correlation between the luminosity of variable stars with their period [Leavitt 1908]. In her 1912 paper, Leavitt found that there was a direct relationship between magnitude and the logarithm of the period. Furthermore, this relationship was defined to include that the apparent magnitude of a star is logarithmically proportional to the absolute magnitude divided

by a distance. [Leavitt 1912] This relationship yields what we now call the distance modulus formula.

$$m - M = 5\log_{10}(d) - 5 \tag{1}$$

The period-luminosity relationship is used for cepheids in distant galaxies. Astronomers calculate the distance to that galaxy using the cepheid's observed period, an absolute magnitude can be calculated. Comparing this to the apparent magnitude, one can estimate the distance to the cepheid and therefore the galaxy using the distance modulus formula.

The flux density refers to the amount of light an observer at some distance receives at given wavelength range.

$$f_\lambda = \frac{L_\lambda}{4\pi r^2} \tag{2}$$

When determining the apparent magnitude of a star in reference to another star, the stellar magnitude equation ,or Pogson's equation, is useful in conjunction with the ratio of flux densities.

$$m_1 - m_2 = -2.5\log_{10}(\frac{f_1}{f_2}) \tag{3}$$

Where luminosity is measured within a range of wavelengths. flux density is measured in the V-band, an optical filter measuring the visual light range of wavelengths.

In this lab report, I measure the apparent magnitude and flux density of RU Cam. To do so, I compared it with a known standard star, Gamma Ursae Majoris. γ Ursae Majoris, or Phecda, is a star in the collestellation of Ursa Major, according to the SIMBAD astronomical database. It has a magnitude of 2.438. [Wenger 2002]

This report also identifies two key sources of uncertainty: photon noise and readout noise. Photon noise is the intrinsic variation associated with photons arriving at the detector. It arises from the random nature of photon emission and is often referred to as "photon shot noise," as it introduces aberrations to the final data point. [Merchant et al. 2022] Readout noise is a limitation on CCDs associated with the accumulation of charge within each pixel. This affects the number of analog to digital units recorded in the final measurement, introducing noise. [Kaaret 2018]

2 Methods

I obtained data using the Boston University Department of Astronomy's 14" telescope in the Judson B. Coit Observatory. The telescope is a Celestron Schmidt-Cassegrain catadioptric model with a primary diameter of 14.25". Along with the telescope, I used MaximDL imaging software to capture and save .FITS files of stellar images and calibration frames. Additionally, the telescope uses TheSky software to remotely operate and point the telescope from a control room. This assisted us when finding and centering the telescope on RU Cam.

I visited the facilities on the nights of February 11, March 2, and March 4, 2026 to observe RU Cam and take the following calibration images: flat frames, bias frames, and dark frames. Each calibration image refined the image to show the stars more clearly by removing the imperfections. To take a bias frame, we put a lens cap on the telescope and set the exposure time to "bias" which was as short as possible in MaximDL. To take a flat frame, we positioned the telescope to focus on a white poster board inside of the observatory dome. We held a light to the poster board, illuminating the white page until the image was half-saturated. We changed the voltage of the lightbulb until the image was half-saturated in MaximDL. We took three sets of bias frames and flat frames, setting the telescope filters to those respective modes. We also took three sets of dark frames but they were ultimately not used in my analysis. In Python, I averaged together the three images of each type

of calibration to create a master bias .FITS file and a master flat .FITS file. Along with the science images and calibration frames, we also imaged Phecda for each of the three nights.

I used Python to analyze and refine the image files as well as SAO DS9 software. I researched the characteristics, right ascension, and declination of RU Cam in the SIMBAD astronomical database. [Wenger 2002] I used Python to analyze the images for values of electrons per ADU which was essential for computing the uncertainty because it showed the ADUs measured by the CCD into photoelectrons. All computations of uncertainty and image analysis were done in Python.

3 Analysis

The goal of this report and the following written code is to determine the magnitude and flux density of RU Camelopardalis on each of the three nights using the images I took with the 14" telescope. A secondary goal was to ascertain the uncertainty associated with photon noise and readout noise from the images.

The objective was to produce two scatter plots depicting the magnitude and flux densities with respect to time and their associated uncertainties.

In a Jupyter Notebook file, I installed the photutils package, a Python library useful for astrophotometry. I also imported other useful libraries such as numpy, astropy, and matplotlib. I compiled all of the data into a folder where each .FITS file was labeled with its exposure, target (e.g. Phecda, RU Cam, etc.), and/or calibration frame type. After each image was read in and averaged using the aforementioned method, I corrected the RU CAM image using the following formula.

$$correctedimage = \frac{masterRUCam - masterbias}{masterflats - masterbias} \quad (4)$$

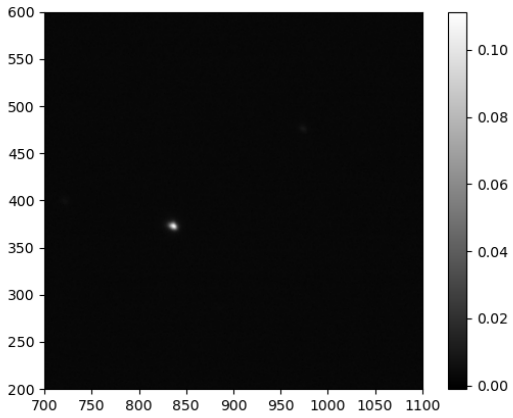


Figure 1: Corrected image of RU Cam on February 11, 2026

	RU Cam	Phecda
Feb. 11	(838, 373)	(675, 311)
Mar. 2	(937, 411)	(485, 297)
Mar. 4	(949, 611)	(781, 411)

Table 1: Pixel coordinates (x,y)

This data was saved to a separate .FITS file and was imported into SAO DS9 to affirm that the image was clean and the star was clearly visible. I repeated this process for Phecda, substituting the RU Cam image for a master standard star image.

From here, I used SAO DS9 to find the pixel coordinates for RU Cam and Phecda. See Table 1 for the pixel coordinates for each night.

In the next line, I used these positions to write an approximation of the circular aperture and annulus – to image and subtract background noise. The aperture radius for this analysis was set to r=12.0. The annulus radii were set to r(in)=15 and r(out)=25. Each was merged with the star images and normalized, yielding the "counts," or electrons detected by a pixel in a charge-coupled device (CCD). The standard star, Phecda, provided a baseline to compare to RU Cam for subsequent calculations. Using ESO data, I determined that the

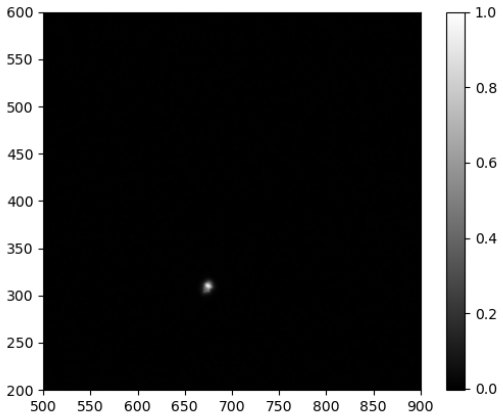


Figure 2: Corrected image of Phecda on February 11, 2026

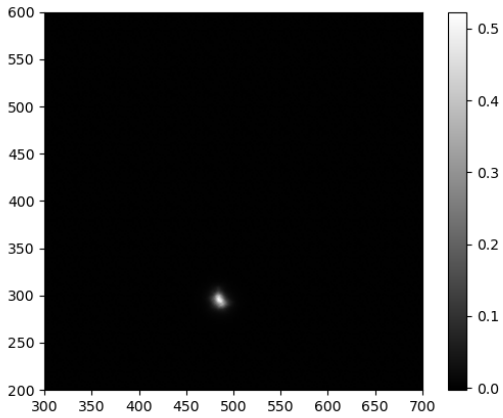


Figure 4: Corrected image of Phecda on March 2, 2026

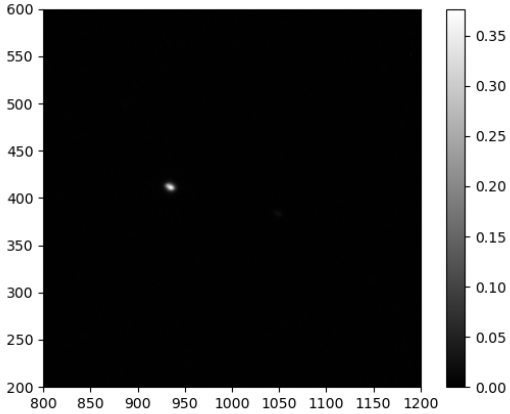


Figure 3: Corrected image of RU Cam on March 2, 2026

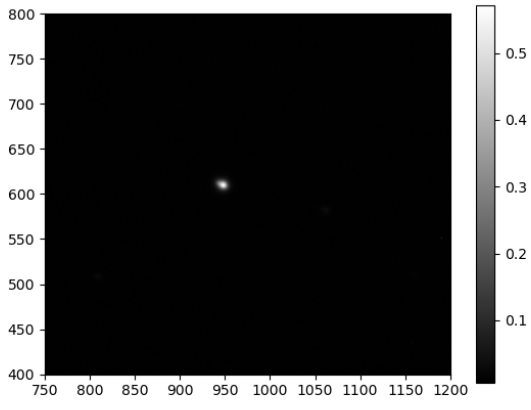


Figure 5: Corrected image of RU Cam on March 4, 2026

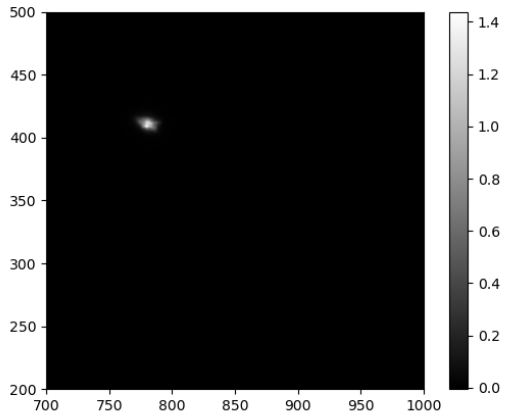


Figure 6: Corrected image of Phecda on March 4, 2026

apparent magnitude of Phecda in the V-band is 2.438. Using the stellar magnitude equation, I calculated the magnitude of RU Cam with respect to the ratio of counts per second, correlating to the exposure time.[1]

The zero point flux density of the V-band is $3.64 \times 10^{-23} Wm^{-2}Hz^{-1}$ I calculated the flux density of the standard star using the following relationship:

$$m = -2.5 \log_{10}(\frac{f}{f_0}) \tag{5}$$

After finding the flux density of Phecda, I calculated the flux density of RU Cam with respect to the ratio of counts per second for the stars.

$$Fv_{RUCam} = Fv_{Phecda} \times \frac{Counts/Sec_{RUCam}}{Counts/Sec_{Phecda}}$$

These operations were repeated for each night. Following that analysis, I translated the dates of each night into their Julian Date. I also created an array that contained the previously calculated magnitudes for each night and an array of the same characteristics for flux density.

[1]See attached code

Before these values could be plotted with respect to time, I had to estimate the uncertainty associated with readout noise and photon noise. To do so, I took the differences between two bias frames.

$$Difference_{N1} = masterbiasN1 - masterbiasN2$$

$$Difference_{N2} = masterbiasN2 - masterbiasN3$$

$$Difference_{N3} = masterbiasN3 - masterbiasN2$$

I took the root mean square of difference to find the initial readout noise with respect to ADUs. In order to convert that to noise with respect to photoelectrons, I multiplied by the gain which was 11.83 for the CCD. I merged the readout noise values per night into a numpy array for easy graphing later.

For the photon noise, I took the square root of the number of counts of the RU Cam images. I also merged these into a numpy array.

To compute the final uncertainty array, I added the two values in quadrature using the equation below.

$$uncertainty = \sqrt{(readout)^2 + (photon)^2}$$

To apply the uncertainty to the magnitude plot, I had to scale the uncertainty values accordingly using the following equation.

$$Error_{magnitude} = 1.0857 \times (\frac{uncertainty}{Counts_{RUCam}})$$

To apply the uncertainty to the flux density, I scaled the uncertainty values accordingly using the equation below.

$$Error_{F_v} = (\frac{uncertainty}{Counts_{RUCam}}) \times F_v$$

Setting time against magnitude, I used the matplotlib "scatter" function to create the scatterplot with data values for each night along with their uncertainty. This plot is displayed in

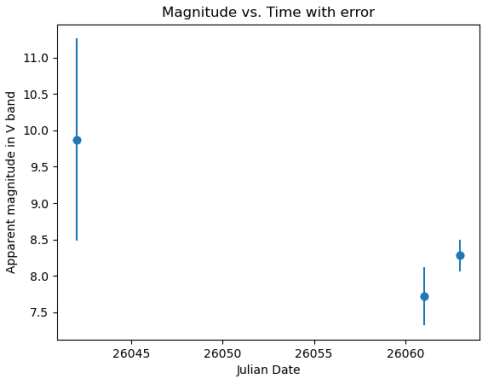


Figure 7: Magnitude of RU Cam

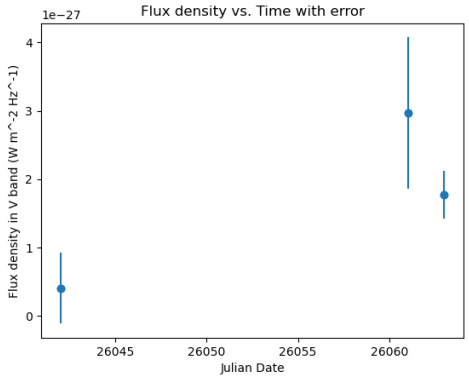


Figure 8: Flux density of RU Cam

Figure 7.

Setting time against flux density in the V band, I used the matplotlib "scatter" function to create the scatterplot with data values for each night along with their uncertainty. This plot is displayed in Figure 8.

4 Discussion

Results

The apparent magnitudes of RU Cam according to the images taken across 21 days with three nights of observation is displayed in Table 2. The accepted magnitude of RU Camelopardalis is between 8.5 and 9.5 in the V-band. [Zaitseva 1973] The flux densities of RU Cam is also denoted in Table 2. There is no flux density that I could find in literature to compare the calculated values.

	Magnitude	Flux Density (Wm^{-2} Hz^{-1})
Feb. 11	9.87	4.085×10^{-28}
Mar. 2	7.72	2.97×10^{-27}
Mar. 4	8.28	1.77×10^{-27}

Table 2: Magnitude and Flux Density of RU Cam

	Magnitude Error	Flux Density Error ($Wm^{-2}Hz^{-1}$)
Feb. 11	1.386	5.217×10^{-28}
Mar. 2	0.404	1.106×10^{-27}
Mar. 4	0.216	3.520×10^{-27}

Table 3: Magnitude and Flux Density Errors

The average value of the three bias exposure frames is 240.8 Analog-to-Digital Units (ADUs). The average value of the three dark exposure frames is 241 ADUs. The 14" telescope has a 12-bit resolution, therefore its saturation limit is 4,095 ADUs.

Astronomers measure the flux density of a star to determine the amount light an observer receives at a given range of wavelengths. In this case, the V-band for visible light is appropriate. Astronomers measure the apparent magnitude, brightness seen at a given location, to determine a star's luminosity with respect to its distance from the observer. I believe it is worthwhile to note that RU Cam is a variable star at an unknown point in its period. Its magnitude fluctuates between 8.5 and 9.5 over 22 days. [Zaitseva 1973] Given that the measured values from this report oscillate between 7.72 and 9.87 across 21 days, it's reasonable to presume that RU Cam may have completed a cycle of variability during observation time.

Explanation of Error

The margin of error could be attributed to multiple causes. The 14" telescope is a ground-based telescope, therefore Earth's atmosphere

could blur the image. Light is distorted by atmospheric particles which makes the star look blurry, making it difficult to pinpoint its exact pixel location in SAO DS9. Additionally, these images were taken at night in Boston, a large city with significant light pollution. Further, error could also be attributed to user error with calibration frames, specifically with imaging flat frames. We took dome flats which introduces error because the white panel might not have been illuminated evenly and not caught the lens's imperfections – adding to some uncertainty within the final corrections.

5 Conclusion

This report concludes that RU Cam is a low magnitude Type II Cepheid variable star. The original goal was to determine the apparent magnitudes and flux densities of this celestial body using three nights of observation data. In the event I repeat this experiment, I would introduce more data points by including more observation nights. Flux density is important to measure because it allows scientists to determine characteristics of the star like distance, temperature, and luminosity.

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